

Distributed Sensitivity To Semantics And Structure Across The Brain During Naturalistic Conversations

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Abstract

The way we use natural language to communicate in everyday conversations is dynamic, contextual, and interactive. However, most evidence on how semantic content and morphosyntactic structure, two core linguistic components, are processed and represented in the brain, comes from controlled comprehension experiments. Therefore, it is unclear how the brain processes these features in real-world settings.

Here, we study how these two linguistic components contribute to neural encoding during both speech production and comprehension in fully naturalistic dialogue. We have analyzed approximately 100 hours of spontaneous, open-ended conversations (over 500,000 spoken words), recorded from four epilepsy patients during multi-day hospital stays, together with high-gamma ECoG activity of 676 intracranial electrodes.

We used time-resolved encoding models to predict lagged neural responses around word onset with the joint representation of the onset-word, and the retro- and prospective context. To probe sensitivity to semantics and structure, we kept the neural data fixed and systematically perturbed the linguistic representations of the context. We compared encoding performance of LLM-derived embeddings of different rephrasing conditions. In both comprehension and production, perturbations targeting semantics and/or structure reduced encoding performance across multiple regions, including the IFG, MTG and TP. At the same time, the relative impact of semantic versus structural perturbations differed across ROIs. For instance, the STG showed stronger sensitivity to structural perturbations in comprehension, and the TP in production was more affected by structural changes. These results are consistent with a broadly distributed sensitivity to semantics and structure, rather than strict regional specialization, and suggest graded regional differences within a shared human neural language network.

Keywords: ECoG, Encoding, Linguistic features, Naturalistic conversations

Background

We use language as a means of interaction: to transfer ideas and express our mental states. This essential form of communication relies on the "mechanics" of language production and comprehension: semantic content and morphosyntactic structure with which we construct our phrases determine how meaning is conveyed and constructed by our brains.

Natural language use in everyday conversation is dynamic, contextual, and fundamentally interactive. Yet most evidence on how these two features, semantic content and morphosyntactic structure, are represented in the brain comes from controlled comprehension experiments using isolated stimuli. These studies suggest that semantics and structure are processed within a broadly shared language network rather than in fully dissociable modules, while also pointing to graded regional differences in their relative contribution (Caucheteux et al., 2021; Fedorenko et al., 2020, 2024; Shain et al., 2024).

However, it remains unclear whether these patterns generalize to real-world dialogue, where language unfolds continuously across longer contexts, is shaped by turn-taking and discourse history, and recruits both comprehension and production processes (Goldstein et al., 2025; Hasson et al., 2012; Zada et al., 2024).

Here, we ask how semantic and structural information contribute to neural encoding during both speech comprehension and speech production in open-ended dialogue in naturalistic conversations, and whether their relative impact differs across brain regions within the broader human neural language network.

Methods

Data and participants

We analyzed approximately 100 hours of spontaneous, open-ended conversations (over 500,000 spoken words), recorded from epilepsy patients (N=4, ages: 24-53) during multi-day hospital stays, together with high-gamma (75-200Hz) electrocorticography (ECoG) activity of 676 intracranial electrodes,



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Brainmap of relative sensitivity to semantic and structural changes

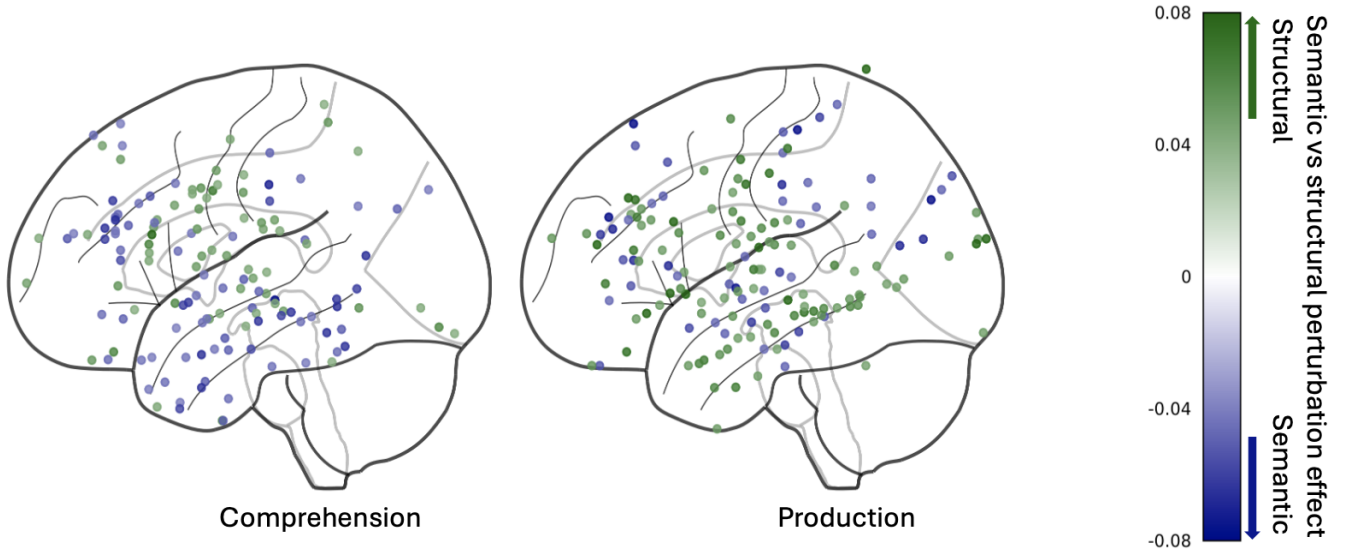


Figure 1: Brainmap of the encoding differences (joint banded ridge regression) between the condition with semantic changes, and the condition with structural changes. $\delta(Enc_{max}(\text{Semantic}), Enc_{max}(\text{Structural}))$, where Enc_{max} = Pearson’s r of the peri-onset timewindow (−500 ms - 500 ms). Intuitively, green: more sensitive to structural changes, blue: more sensitive to semantic changes. Electrodes were thresholded to increase interpretability of the figure; comprehension ($\delta Enc_{max} > 0.03$, $N=266$) and production ($\delta Enc_{max} > 0.035$, $N=288$).

which was z-scored and smoothed with a 50-ms Hamming kernel (Goldstein et al., 2025).

Model training and evaluation

We used time-resolved encoding models to predict lagged neural responses around word onset (−500 ms to 500 ms) with the joint representation of the onset word, and the retro- and prospective context, which were operationalized with the eight words prior to, and following the onset word, respectively. We used Banded Ridge models (Dupre la Tour et al., 2022), which were cross-validated on 90/10% conversation-based train/test splits. To probe sensitivity to semantics and structure, we kept the neural data fixed and systematically perturbed the linguistic representations of the context by rephrasing the conversations. Semantic sensitivity was probed by replacing all content words with random alternatives that preserve structural scaffolding while disrupting lexico-compositional meaning; structural sensitivity was probed by paraphrasing the context, such that meaning is preserved, while altering morphosyntactic form. We compared encoding performance of LLM-derived embeddings of different rephrasing conditions.

Results

In both comprehension and production, perturbations targeting semantics and/or structure reduced encoding performance across multiple canonical language net-

work regions, including the inferior frontal gyrus, the middle temporal gyrus and the temporal pole. Additionally, in most brain regions, the combined manipulation of semantics and structure was more disruptive than either manipulation alone.

At the same time, the relative impact of semantic versus structural perturbations differed across regions. In comprehension, the superior temporal and fusiform gyri showed stronger sensitivity to structural than semantic perturbations, while the other ROIs showed greater semantic sensitivity. In production, areas such as the temporal pole and pre-central gyrus showed stronger sensitivity to structural perturbations, whereas, for instance, the rostral middle frontal and superior frontal gyri were more sensitive to semantic changes.

Conclusion

Together, these findings suggest that both semantic and structural information contribute to neural encoding across a distributed language network during naturalistic conversations, in both comprehension and production. Rather than indicating categorical functional specialization, the results support a shared network with graded regional differences in relative sensitivity to these linguistic features. This extends prior work from controlled comprehension paradigms and highlights naturalistic conversation as an important setting for studying how the brain processes language in the real-world.

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